TEXAS INSTRUMENTS: TECHNICAL TEST

Date: 20th December 2003

1. For a CMOS inverter, the transition slope of Vout vs Vin DC characteristics can be increased (steeper transition) by:

a. Increasing W/L of PMOS transistor
b. Increasing W/L of NMOS transistor
c. Increasing W/L of both transistors by the same factor
d. Decreasing W/L of both transistor by the same factor

Ans: c

2. Minimum number of 2-input NAND gates that will be required to implement the function: Y = AB + CD + EF is

a. 4
b. 5
c. 6
d. 7

ans: c

3. Consider a two-level memory hierarchy system M1 & M2. M1 is accessed first and on miss M2 is accessed. The access of M1 is 2 nanoseconds and the miss penalty (the time to get the data from M2 in case of a miss) is 100 nanoseconds. The probability that a valid data is found in M1 is 0.97. The average memory access time is:

a. 4.94 nanoseconds
b. 3.06 nanoseconds
c. 5.00 nanoseconds
d. 5.06 nanoseconds

ans: a

4. Interrupt latency is the time elapsed between:

a. Occurrence of an interrupt and its detection by the CPU
b. Assertion of an interrupt and the start of the associated ISR
c. Assertion of an interrupt and the completion of the associated ISR
d. Start and completion of associated ISR

Ans: d (not confirmed)

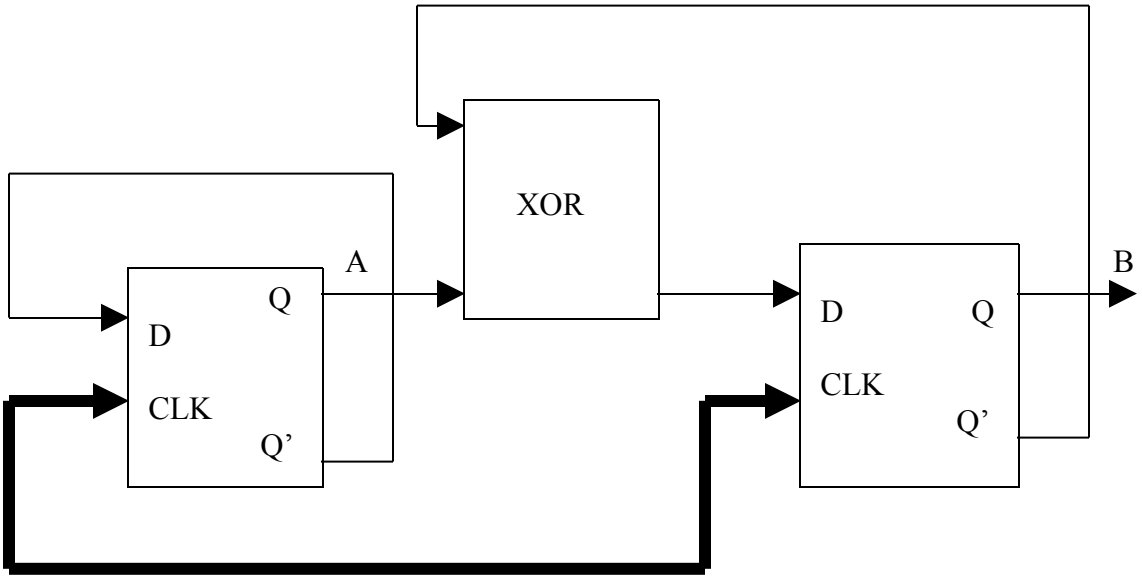
5. Which of the following is true for the function (A.B + A'.C + B.C)

a. This function can glitch and can be further reduced
b. This function can neither glitch nor can be further reduced
c. This function can glitch and cannot be further reduced
d. This function cannot glitch but can be further reduced

Ans: c **This can be reduced further using K-map, don't know abt glich, but it should not glitch**

6. For the two flip-flop configuration below, what is the relationship of the output at B to the clock frequency?

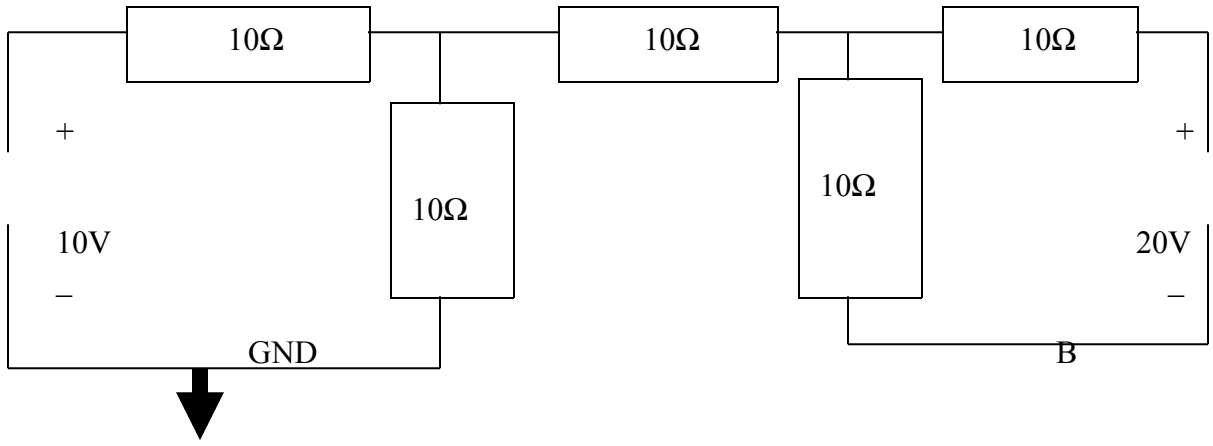
a. Output frequency is 1/4th the clock frequency, with 50% duty cycle
b. Output frequency is 1/3rd the clock frequency, with 50% duty cycle
c. Output frequency is 1/4th the clock frequency, with 25% duty cycle
d. Output frequency is equal to the clock frequency



Ans: a

7. The voltage on Node B is:

a. 0
b. 10
c. –10
d. –5



Ans: d

8. A CPU supports 250 instructions. Each instruction op-code has these fields:

- The instruction type (one among 250)
- A conditional register specification
- 3 register operands
- Addressing mode specification for both source operands

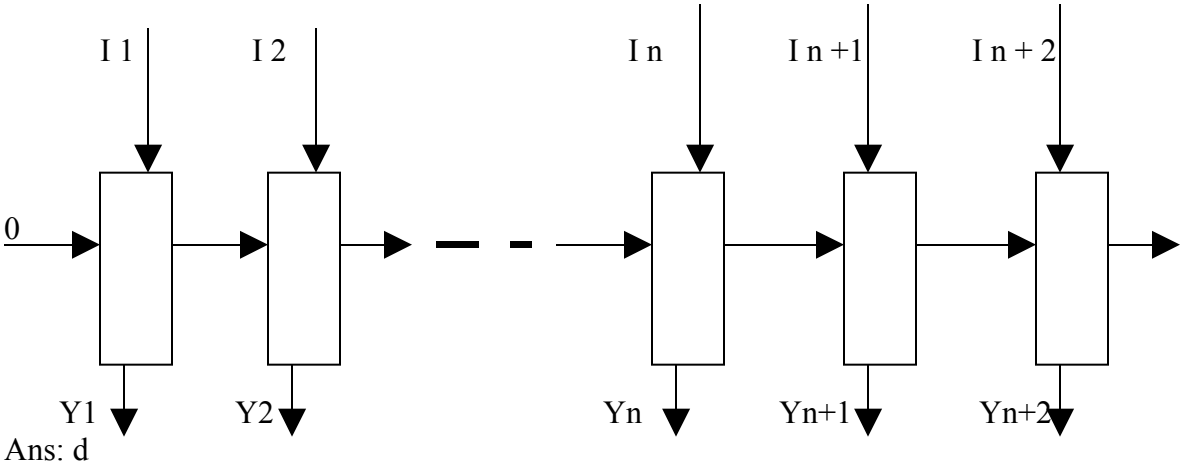
The CPU has 16 registers and supports 5 addressing modes. What is the instruction op-code length in bits?

a. 32
b. 24
c. 30
d. 36

ans: don't know

9. In the iterative network shown, the output Yn of any stage N is 1 if the total number of 1s at the inputs starting from the first stage to the Nth stage is odd. (Each identical box in the iterative network has two inputs and two outputs). The optimal logic structure for the box consists of:

a. One AND gate and one NOR gate
b. One NOR gate and one NAND gate
c. Two XNOR gates
d. One XOR gate



Ans: d

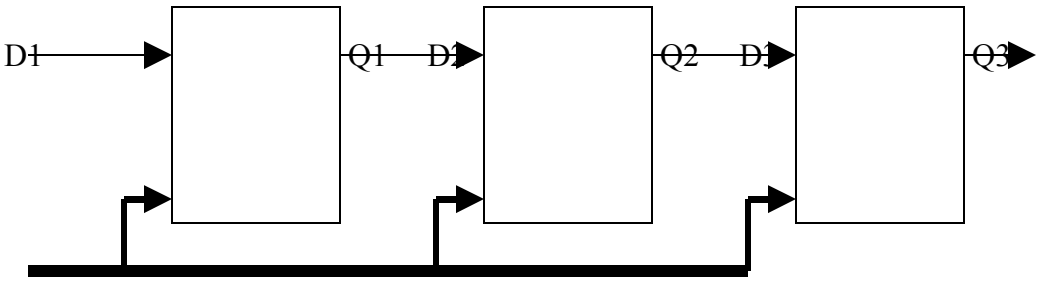
10. Consider a circuit with N logic nets. If each net can be stuck-at either values 0 and 1, in how many ways can the circuit be faulty such that only one net in it can be faulty, and such that up-to all nets in it can be faulty?

a. 2 and 2N
b. N and 2^N
c. 2N and 3^N-1
d. 2N and 3N

ans: 2N and 2^N (no match) see it .

sorry , no idea abt this

11. In the circuit shown, all the flip-flops are identical. If the set-up time is 2 ns, clock->Q delay is 3 ns and hold time is 1 ns, what is the maximum frequency of operation for the circuit?



a. 200 MHz
b. 333 MHz
c. 250 MHz
d. None of the above

Ans: a

12. Which of the following statements is/are true?

I. Combinational circuits may have feedback, sequential circuits do not.
II. Combinational circuits have a 'memory-less' property, sequential circuits do not.
III. Both combinational and sequential circuits must be controlled by an external clock.

a. I only
b. II and III only
c. I and II only
d. II only

Ans: d

13. Consider an alternate binary number representation scheme, wherein the number of ones M, in a word of N bits, is always the same. This scheme is called the M-out-of-N coding scheme. If M=N/2, and N=8, what is the efficiency of this coding scheme as against the regular binary number representation scheme? (As a hint, consider that the number of unique words represent able in the latter representation with N bits is 2^N. Hence the efficiency is 100%)

a. Close to 30%

b. Close to 50%
c. Close to 70%
d. Close to 100%

Ans: a

14. A CPU supports 4 interrupts- I1, I2, I3 and I4. It supports priority of interrupts. Nested interrupts are allowed if later interrupt is higher priority than previous one. During a certain period of time, we observe the following sequence of entry into and exit from the interrupt service routine:
I1-start---I2-start---I2-end---I4-start---I3-start---I3-end---I4-end---I1-end
From this sequence, what can we infer about the interrupt routines?

a. I3 > I4 > I2 > I1
b. I4 > I3 > I2 > I1
c. I2 > I1; I3 > I4 > I1
d. I2 > I1, I3 > I4 > I2 > I1

Ans: c

15. I decide to build myself a small electric kettle to boil my cup of tea. I need 200 ml of water for my cup of tea. Assuming that typical tap water temperature is 25 C and I want the water boiling in exactly one minute, then what is the wattage required for the heating element?
[Assume: Boiling point of water is 100 C, 1 Calorie (heat required to change 1 gm of water by 1 C)= 4 joules, 1 ml of water weighs 1 gm.]

a. Data given is insufficient
b. 800 W
c. 300 W
d. 1000 W
e. 250 W

ans: d

16. The athletics team from REC Trichy is traveling by train. The train slows down, (but does not halt) at a small wayside station that has a 100 mts long platform. The sprinter (who can run 100 mts in 10 sec) decides to jump down and get a newspaper and some idlis. He jumps out just as his compartment enters the platform and spends 5 secs buying his newspaper that is at the point where he jumped out. He then sprints along the platform to buy idlis that is another 50 mts. He spends another 5 secs buying the idlis. He is now just 50 mts from the other end of the platform where the train is moving out. He begins running in the direction of the train and the only other open door in his train is located 50 mts behind the door from where he jumped. At what(uniform) speed should the train be traveled if he just misses jumping into the open door at the very edge of the platform? Make the following assumptions

- He always runs at his peak speed uniformly
- The train travels at uniform speed
- He does not wait (other than for the idlis & newspaper) or run baclwards

a. Data given is insufficient
b. 4 m/s
c. 5 m/s

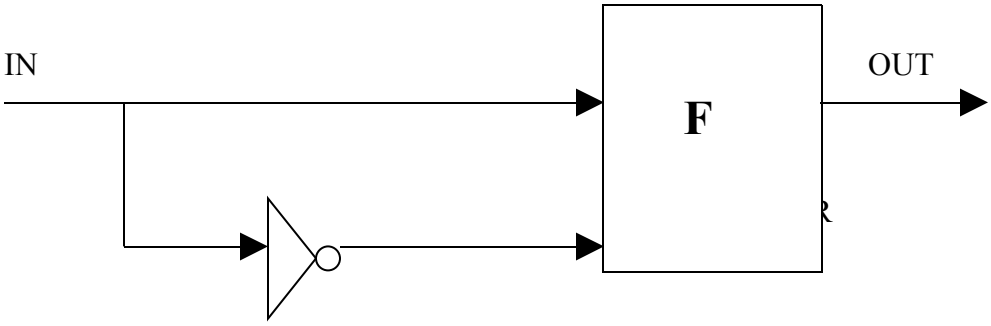
d. 7.5 m/s
e. 10 m/s

ans: c

17. State which of the following gate combinations does not form a universal logic set:
a. 2-input AND + 2-input OR
b. 2-to-1 multiplexer
c. 2-input XOR + inverter
d. 3-input NAND

ans: a

18. For the circuit shown below, what should the function F be, so that it produces an output of the same frequency (function F1), and an output of double the frequency (function F2).

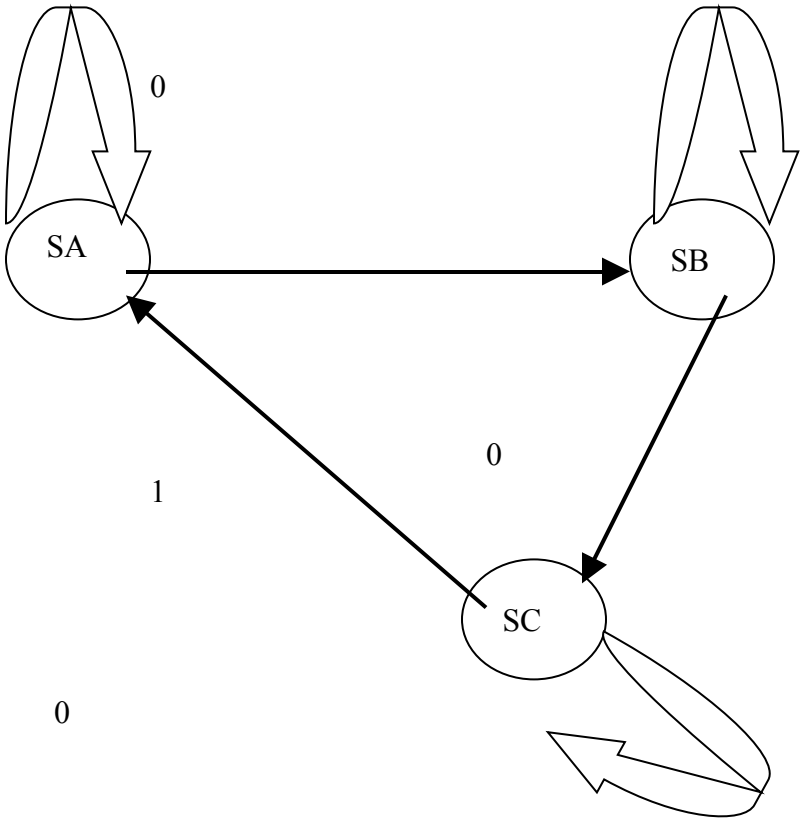


a. F1= NOR gate and F2= OR gate
b. F1=NAND gate and F2= AND gate
c. F1=AND gate and F2=XOR gate
d. None of the above

Ans: c

19. The FSM (finite state machine) below starts in state Sa, which is the reset state, and detects a particular sequence of inputs leading it to state Sc. FSMs have a few characteristics. An autonomous FSM has no inputs. For a Moore FSM, the output depends on the present state alone. For a Mealy FSM, the output depends on the present state as well as the inputs. Which of the statements best describes the FSM below?

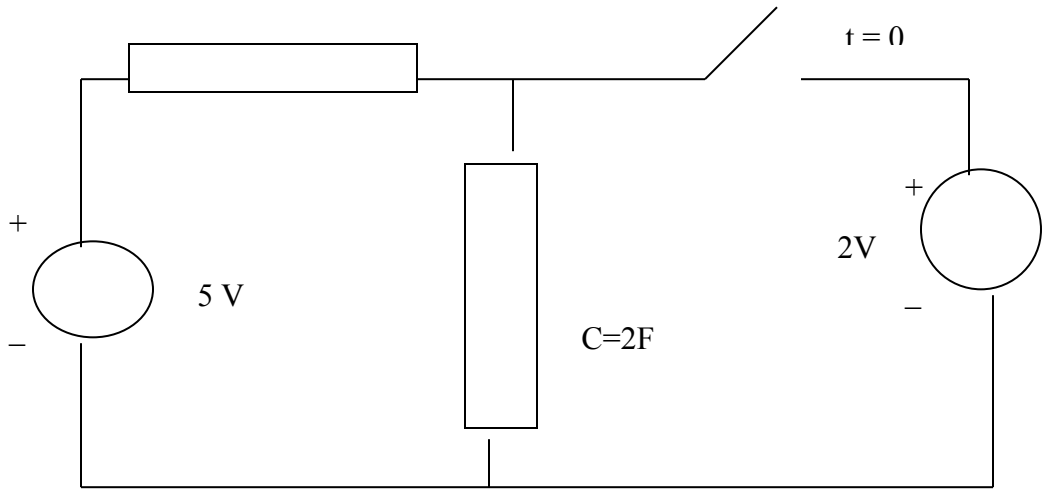
a. It has two states and is autonomous
b. The information available is insufficient
c. It is a Mealy machine with three states
d. It is a Moor machine with three states



Ans :d

20. In the circuit given below, the switch is opened at time t=0. Voltage across the capacitor at t=infinity is:

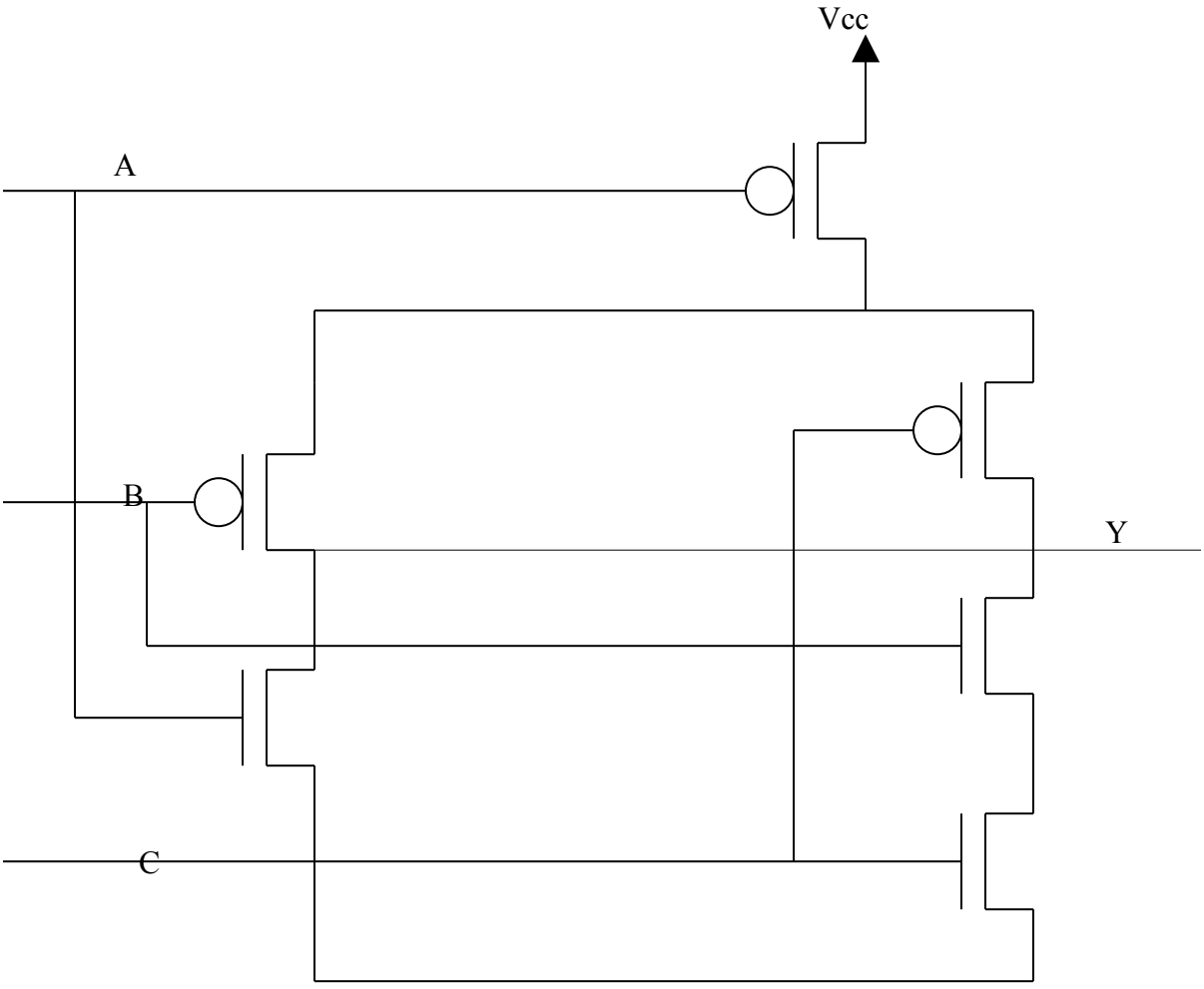
a. 2V
b. 3V
c. 5V
d. 7V



Ans: c

21. What is the functionality represented by the following circuit?

a. y= ! (b+ac)
b. y= ! (a+bc)
c. y= ! (a(b+c))
d. y= ! (a+b+c)



Ans: b

22. The value (0xdeadbeef) needs to stored at address 0x400. Which of the below ways will the memory look like in a big endian machine:

	0x403	0x402	0x401	0x400
a.	be	ef	de	ad
b.	ef	be	ad	de
c.	fe	eb	da	ed
d.	ed	da	eb	fe

ans: don't know

ans should be (b), just check with some CS guy, little endian is Intel type, Big-endian is perhaps Motorola type

23. In a given CPU-memory sub-system, all accesses to the memory take two cycles. Accesses to memories in two consecutive cycles can therefore result in incorrect data transfer. Which of the following access mechanisms guarantees correct data transfer?

a. A read operation followed by a write operation in the next cycle.
b. A write operation followed by a read operation in the next cycle.
c. A NOP between every successive reads & writes
d. None of the above

Ans: c(not confirm)
I'm also not sure.

24. An architecture saves 4 control registers automatically on function entry (and restores them on function return). Save of each registers costs 1 cycle (so does restore). How many cycles are spent in these tasks (save and restore) while running the following un-optimized code with n=5:

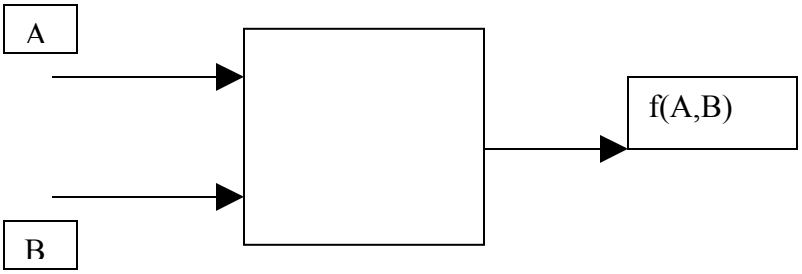
```
Void fib(int n)
{
        if((n==0) || (n==1)) return 1;
        return(fib(n-1) + fib(n-2));
}
```

a. 120
b. 80
c. 125
d. 128

ans: a

25. The maximum number of unique Boolean functions F(A,B), realizable for a two input (A,B) and single output (Z) circuit is:

a. 2
b. 6
c. 8
d. None of the above

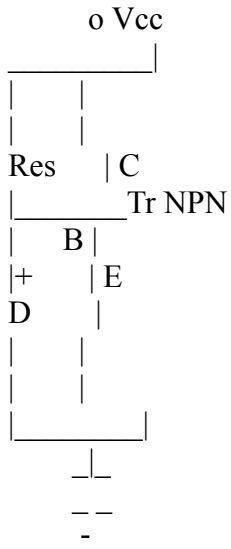


Ans: 2*(2*2)=16 ie d

paper of TI 1999

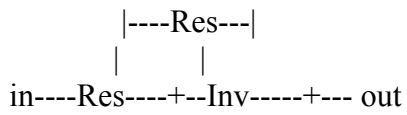
Hard ware part only. There was one part of reasoning and there was separate paper for software persons.

1.



Find the current I delivered by the battery.

2.



What is the given circuit

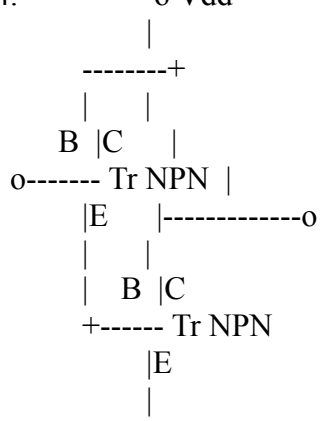
a) Latch b)Amplifier c)Schmitt trigger. d)

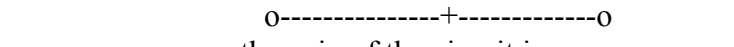
3. The total no of flip flop required for N stage sequential circuit

N N-1 N

a)2 b)2 c) Log N d) 2 -1

4.

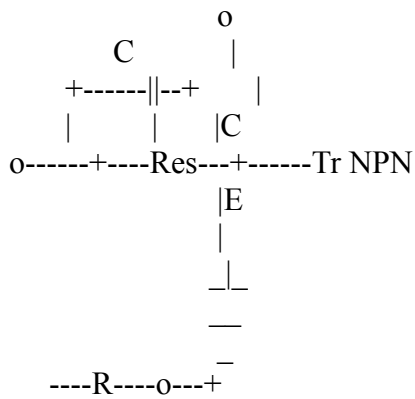




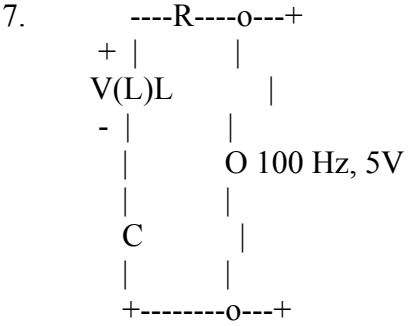
the gain of the circuit is
a) beta square b)beta + 1 c) (beta+1) ka square d)

5. If the o/p and i/p are related by y=k(x square) and i/p is a sum of 2 waveforms then the modulation scheme is
a) FM b)AM c)PM and d)None
Ans. B

6.Function of C in the circuit below is
a) Improve switching b)dc coupling c) ac coupling d) None

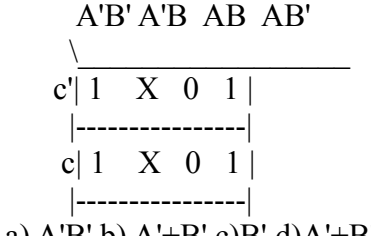


7.



if the ckt is at resonance and V(L)= (constant) V (given)
the value of V(R) and V(C) is
a)100V,5V b)-100V,5V c)5V,5V
(Use V(L)=5 /_100 and V(C)=5/_-100, V(R)=5V

8. Minimize the K-map



a) A'B' b) A'+B' c)B' d)A'+B'+C'

9. IF the rate of removal of elements in a queue containing N elements is proportional to the no of elements already existing in the queue at that

instant then the no. of elements----
a)decrease linearly b)Exponetialy decrease b) Logarithmcally

10. One question on CMOS ckt.
11. Two question on OP-AMP.

THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple ckt.one ^M
>transistor is diode equivalent. & asked the o/p across the 2 nd transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
> Emitter^M
>---R-------transistorbase| --^M
> | ---^M
> collector^M
> in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
> in fig.capacitor is used for in this ckt:^M
>^M
>^M
> ans:a.speedupb.active bypass c.decoupling^M
> 4.^M
>^M
> -----R------I----------o/p^M
> |___R____ |^M
> in above r is resistence.I is cmos inverter.^M
> then ckt is used for:^M
>^M
>^M
> a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
> 5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
> amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
> from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
> 6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M

> inductor is given.R I C values are given & asked for^M
> voltages across resistence & capacitor.^M
> 7.^M
> ___ R_____^M
> | |^M
> ---R------OPAMP ----------^M

> |---^M
> R1 R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
> ground^M
>^M
>^M
> 8.asked for Vo at the o/p.it is like simple cmos realization that is n ^M
>block is above^M
> & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
> 9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay ^M
>is 1 nanosec.^M
> A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
> a.updown^M
> b.up c. updown glitching like that (take care abt glitching word)^M
>^M
> 10.^M
>^M
>^M
> ----------------| subtractor|---------o/p^M
> |___HPF____|^M

>^M
> the ckt is LPF ,HPF or APF ?^M
>^M
> 11.in a queue at the no of elements removed is proportional to no of ^M
>elements in^M
> the queue.then no of elements in the queue:^M
> a.increases decreases exp or linearly(so these are the 4 options given ^M
>choose 1 option)^M
> 12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the ^M
>fastest in the^M
> following implementations.^M
> ans we think ((AB)(CD))((EF)(GH))^M
> 13.with howmany 2:1 MUX u can for 8:1 MUX.answer is 7.^M
> 14. there are n states then ffs used are log n.^M
> 15.cube each side has r units resistence then the resistence across ^M

>diagonal of cube.^M
> 16.op amp connections asked for o/p^M
> the answer is (1+1/n)(v2-v1).check it out.practise this type of model.^M
> 17.^M
> _____ supply^M
> ---|__ ___|^M
> Ii >_____ |___ Tranistot^M
> > _____Vo^M
> > _____Vo^M
> |^M
> |^M
> R |^M
> | | Io^M
> ground.^M
>^M
>^M
>^M
>^M
> asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
> a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
> 18.y=kxsquare. this is transfer function of a block with i/p x & o/p ^M
>y.if i/p is^M
> sum of a & b then o/p is :--^M
>^M
> a. AM b.FM c. PM^M
> 19.^M
> ------MULTIPLIER--- |^M

> | |^M
> _____R__|__OPAMP_____Vo^M
> ---^M
> |^M
> ground.^M
> v in = -Ez then o/p Vo =?^M
> answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
> is a.b;^M

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are
diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c 1 x 0 0
1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}

what's the output .

1.10,5
2,10,10
c.5,5
d. none

5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

7.#define f(a,b) a+b
#define g(a,b) a*b

main()
{

```
int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.

```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(_expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

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1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be

a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c 1 x 0 0
1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}

what's the output .

1.10,5
2,10,10
c.5,5

d. none

5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

7.#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}

what's the value of m
a.70
b.50
c.26
d. 69

8.
```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(_expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time

b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

bye..
p.sreenivasa rao

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_____
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> ATTACHMENT part 5 application/msword name=texas.doc

> ATTACHMENT part 6 application/octet-stream name=texas

Date:
Wed, 30 Dec 1998 19:30:34 +0500
From:
PVSAK Viswanadham Add to Address Book
Subject:
TI
Organization:
Computer Science Dept., Indian Institute of Technology, Kharagpur
To:
bkup

for TI aptitude test consist of all pictorial questions. ie in each question he will give 8 diagrams and ask to find the 9'th diagram in that sequence.
You go through RS Agarwal. These aptitude questins are very easy. Just pratice them. In RS Agarwal gothrough SERIES chapter. It is suffient. There are 35 aptitude questions. First 25 are very easy. Do these questions

in just 15 or 20 minutes. Because last questions are
very touch.

TECHNICAL TEST:
---

1)3 flipflops are connected so that after 0 to 5 count occured
next number is zero. So what is the counter?
Ans: mod 6 counter

2)simplication of some boolean _expression which is simple.
Boolean _Expression is A+A'B.
Ans:A+B

3)Given inorder sequence and preorder sequence and asked to
find out postorder sequence.

4)Some question on value of a static variable.

5) Given an interger in binary form,find the number of ones in
that number without counting each bit.(This questin is not
multiple choice question. This question carries more
marks. So please take care for this question.)

6) 1-way set associative memory is called-----
a)direct b)something c)1-way set associative 4)something
Ans: c

7)Fastest IPC mechanism is

a)shared memory b)pipes c)named pipes d)semaphores
Ans:c

8)Some page references are given. You are asked to implement
it with Least Frequently Used algorithm.

9)Some diagram is given.
Iam describinmg the diagram. A 2*1 MUX is given. The inputs
are A,B. Output is C. C and A are tied together. What is
the diagram.?

Ans:Latch.

****

This paper is for Electrical & Electronics students. There is

separate test for computer Science Students. There are 20 questions.

1)Some circuit is given. Iam describing the circuit.
A resistor R & a capacitor C are connected in parallel.
To this circuit another circuit which is having a capacitor of capacity 2C & an impedence Z, is connected in series.
You are asked to find out the value of Z? Note that 2C & Z are connected in series.
a)Z=2C
b)Z=2L
c)Z=L/2
d)Z=2R

2)Some circuit which consist of only resistors R is given.
This is a repetative circuit. U have to find the effctive resistance of the entire circuit.
A)Rin=R
B)Rin=(5+sqrt(3))/7
C)Rin=(19+sqrt(3))/8
D)None.

3)Two wave forms are given. You are asked to write the cirsuit to get B(second wave form) from A(first wave form).

4)#define SUM(a,b) a+b

```
main()
{
a=2;
b=3;
x=SUM(a,b)*2;
printf("x=%d\n",x);
}
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
}

main()
{
static int i=0;
number(i);
```

}
Ans: I don't know.

6)Some circuit is given. I can't describe the circuit. There are
3 resistors,3 capacitors & one inverter.. The question is
What is the value of the frequency such that the circuit oscillates.
A)f=RC
B)f=sqrt(3)/(Pi*R*C)
C)f=1/(Pi*R*C)
D)something
Ans:I don't know the answer.

7)Question on flipflop. So gothrough all flipflops.

8)There are 5 questions on Nmos & Pmos circuits.

****
This Paper is for Computer Science Students. THis paper is
very easy. You can definitely do it in one hour.
****

(1) The fastest memory is
(i) DRAM, (ii) ROM, (iii) SRAM, (iv) Main memory
Ans : SRAM

(2) Programing exceptions are
(i) Asynchronous, (ii) Synchronous, (iii) None
Ans : Asynchronous

(3) DSP which architecture is used
(i) MIMD, (ii) SIMD, (iii) Nueman, (iv) Harvard Architecture
Ans : Harvard Architecture

(4) C prog. for searching for an element in linked list

(5) main()
{
unsigned char i;
int sum;

for(i=0; i<300; i++)
sum+ = i;
printf("\nSum = %d\n", sum);
}

Ans : infinite loop

(6) void fn(int *p)
{
static int val = 100;
p = &val;
}

main()
{
int i=10;
printf("i=%d\n", i);
fn(&i);
printf("i=%d\n", i);
}

Ans : i=10 i=10

(7) int a[10[15];
char b[10[15];
(a) location g a[3][4], if base location g a[0][0] is ox1000
(b) location g b[3][4], if base location g b[0][0] is ox2000
int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans : A _____
--------|2*1 MUX |
B | |--------o/p
--------| |
| -----------
|_____|C

B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

(i) x = x+y;
y = x-y;
x = x-y;

(ii) x = x^y;

```
y = x^y;
x = x^y;
```

(11) Count no of 1's in a word without using bit by bit.
(This question carries more marks. It is not a multiple choice question.)

(12) Code 1 :

```
for(i=0; i<1000; i++)
for(j=0; j<100; j++)
x = y;
```

Code 2 :

```
for(i=0; i<100; i++)
for(j=0; j<1000; j++)
x = y;
```

Which code will execute faster
(i) Code 1 and Code 2 are of same speed,
(ii) Code 1,
(iii) Code 2,
(iv) None.

Ans : Code 2

(13)

```
main()
{
int a[10] = {1, 2, 3, ...., 10}, i, x=10, temp;

for(i=0; i temp = a[i];
a[i] = a[x-i-1];
a[x-i-1] = temp;
}
```

(i) All contents of array a are reversed
(ii) Only some portions are altered
(iii) Remains same
(iv) None

Ans : (iii)

(14) An array is stored in row major order. The memory capacity is 30 MB. And in unix system demand paging is used. Which one will give more page faults?

```
#define V_L_I 10000
int i, j, array[V_L_I][V_L_I];
```

Code 1 :
array[i][j] = 1;

Code 1 :
for(j=0; j for(i=0; i array[i][j] = 1;

Ans : Code 2

(15) In C which parameter passing technique is used?
(i) call by value,
(ii) call by reference,
(iii) both

Ans : call by value

(16) A circuit is given with 2 exclusive OR gates whose boolean
_expression will be y = '(AB) + AB
(' indicates bar)

(17) main()
{
int i = 1;
fork();
fork();
printf("\ni = %d\n", i+1);
}

Ans : 4 printfs will occur and i = 2

(18) Compute the complexity of Binary search.
Ans : O(lg n) (Answer in detail. This is not a multiple choice question.
It carries more marks.)

(19) Write _expression for the tree graph :
Ans : ((a-b) + c*d)/x

(20) # define MAX(a, b) a>b ? a:b
main()
{
int m, n;
m = 3 + MAX(2, 3);
n = 2 * MAX(3, 2);
printf("m = %d, n = %d\n", m, n)
}

Ans : m=2, n=3
paper of texas instruments.
Technical + aptitude + interview.
* Questions on c
here i am not strictly following syntax it is just to show what was asked.
1 #define sum(a,b) a+b
value of sum(2,3)*2 Ans:8

2. a=5,b=6
swap(&a,b);
This function is written to swap a and b
find value of a and b . Ans 6,6

3.
```
function()
{
static int a=0;
a=a+1;
return a;
}
main()
{
function();
function();
function();
printf a;
}
```
final value of a ? Ans : a=3. static initializes once.
4.Write two prog. to swap a & b without using temp variable.
5.
```
unsigned char i;
int sum=0;
for(i=0;i<300;i++)
sum=sum+i;
printf(sum);
```

Ans:Program will held in infinite loop b/c i can not exceed 255.

6.
five questions on MOSFETS.
four were having single mosfets.
questions were simple.as i told u in Gwalior.
just on the funda that it will conduct if Vg-Vs > Vt .

In one question output at drain was to be calculated while o/p was initially charged to 5v and to the gate 5v were applied.

In one question output at drain was to be calculated while o/p was initially charged to 5v and the gate was shorted to drain.
7.
Clear the concept of settling time , hold time and other times. 3 ques on that.
like values of various delays were given and max frequency at which the circuit can work
hint : 1/sum of all delays . In our case ans was 200 Mhz.

8. An input and output waveform was given and circuit was to be designed with the use of one
delay.
Ans : exor gate in which second input is first input with a delay.

9. A question to determine sequence of counter. Don't get puzzled it was a tough question.

10.The output and input of a inverter is connected by three RC stages in between of each stage two amplifiers with poles at imaginary axis were connected.
Hint : The poles at imaginary axis will create extra 180 phase shift thus the circuit will oscillate and calculate the frequency of operation.

11. A series of infinite connected rc circuit and overall input resistance is calculated.(question of 12 class)

```
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        -------------------------

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          next number is zero. So what is the counter?
          Ans: mod 6 counter

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to get B(second wave form) from A(first wave form).

4)
```
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  x=SUM(a,b)*2;
  printf("x=%d\n",x);
  }
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
```

```
}

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Ans: I don't know.

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(4) C prog. for searching for an element in linked list

(5)
```
main()
{
        unsigned char   i;
        int     sum;

        for(i=0; i<300; i++)
                sum+ = i;
        printf("\nSum = %d\n", sum);
}
```

Ans : infinite loop

(6)
```
void fn(int *p)
        {
                static int val = 100;
                p = &val;
```

```
        }

main()
{
        int i=10;
        printf("i=%d\n", i);
        fn(&i);
        printf("i=%d\n", i);
}
```

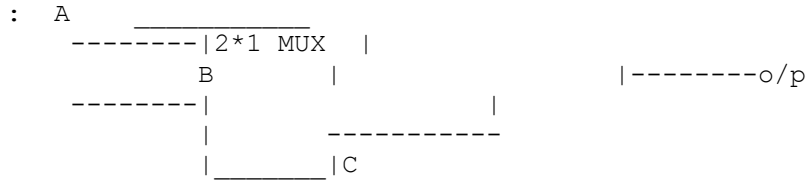
Ans : i=10 i=10

(7) int a[10[15];
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(a) location g a[3][4], if base location g a[0][0] is ox1000
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int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans :



B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

(i)
```
x = x+y;
y = x-y;
x = x-y;
```

(ii)
```
x = x^y;
 y = x^y;
 x = x^y;
```

(11) Count no of 1's in a word without using bit by bit.

>^M
> THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple ckt.one ^M

>transistor is diode equivalent. & asked the o/p across the 2 nd transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
>   Emitter^M
>---R-------transistorbase| --^M
> | ---^M
>   collector^M
>   in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
>   in fig.capacitor is used for in this ckt:^M
>^M
>^M
>   ans:a.speedupb.active bypass c.decoupling^M
>   4.^M
>^M
>   -----R------I----------o/p^M
> |___R____ |^M
>   in above r is resistence.I is cmos inverter.^M
>   then ckt is used for:^M
>^M
>^M
>   a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
>   5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
>   amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
>   from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
>   6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M
>   inductor is given.R I C values are given & asked for^M
>   voltages across resistence & capacitor.^M
>   7.^M
>   ___ R_____^M
> | |^M
>   ---R------OPAMP ----------^M

                  > |---^M
>   R1   R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
>   ground^M
>^M
>^M

>   8.asked for Vo at the o/p.it is like simple cmos realization that is n ^M
>block is above^M
>   & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
>   9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay ^M
>is 1 nanosec.^M
>   A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
>   a.updown^M
>   b.up c. updown glitching like that (take care abt glitching word)^M
>^M
>   10.^M
>^M
>^M
>   ----------------| subtractor|---------o/p^M
> |___HPF____|^M

                        >^M
>   the ckt is LPF ,HPF or APF ?^M
>^M
>   11.in a queue at the no of elements removed is proportional to no of ^M
>elements in^M
>   the queue.then no of elements in the queue:^M
>   a.increases decreases exp or linearly(so these are the 4 options given ^M
>choose 1 option)^M
>   12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the ^M
>fastest in the^M
>   following implementations.^M
>   ans we think ((AB)(CD))((EF)(GH))^M
>   13.with howmany 2:1 MUX u can for   8:1 MUX.answer is 7.^M
>   14. there are n states then ffs used are log n.^M
>   15.cube each side has r units resistence then the resistence across ^M
>diagonal of cube.^M
>   16.op amp connections asked for o/p^M
>   the answer is (1+1/n)(v2-v1).check it out.practise this type of model.^M
>   17.^M
>   _____________ supply^M
>   ---|__   ___|^M
> Ii   >________ |___   Tranistot^M
>   > _______Vo^M
>   > _______Vo^M
> |^M
> |^M
>   R |^M
> | | Io^M
>   ground.^M
>^M
>^M
>^M

>^M
>   asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
>   a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
>   18.y=kxsquare. this is transfer function of a block with i/p x & o/p ^M
>y.if i/p is^M
>   sum of a & b then o/p is :--^M
>^M
>   a. AM b.FM c. PM^M
>   19.^M
>   ------MULTIPLIER--- |^M

                       > | |^M
>   _____R__|__OPAMP_____________________Vo^M
>   ---^M
> |^M
>   ground.^M
>   v in = -Ez then o/p Vo =?^M
>   answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
>   is a.b;^M

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c   1   x 0 0
       1   x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]

IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

```
4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```
what's the purpose of the program

a. its output is hex representation of i

b. bcd
c. binary
d. decimal

7.#define f(a,b) a+b
#define g(a,b) a*b

```
main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
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```
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d. size of a can be same if ...

Hi friends

here is the full paper of TI India IITB 20/07/2001.
WITH SOME ANSWER

the paper had 4 sections in which we have to attend
2 sections
i attempted digtal and analog

and in apti, there was 75 Qs in 60min. be prepare
for
it also, apti was very tough. Apti was more
mathematical stuff, requires a
lot of thinking and very high speed. ex. coding,
reading compreh.(RC),
calculations, relationship, travelling problem
(given some cities and
journey conditions were given) etc etc...

Best of luck
Regards
Vijay Mathur

DIGITAL

1. nand gate is
a) associative &cumulative b)cumulative but not associative
c)not cumulative but associative d)not cumultive and associative

ANS. b

2. which imp has les delay
a) (a xor b) xor (c xor d)
b) (((a xor b) xor c) xor d)
(think on the situation when input a, b, c, d are comes in ascending (i.e. a frist then b then c and then d) and desending (opposite order)

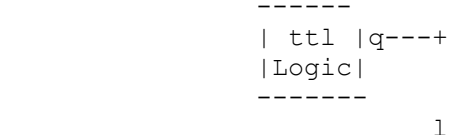
3) one inverter cmos circuit was given with A variable and enable B signal

ANS. tristate inverter with B as enable

4)a logic cell which dertermines(op =1) for odd no.. of 1s in the given seq is

ANS. one xor gate

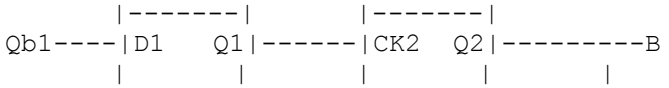
5)circuit

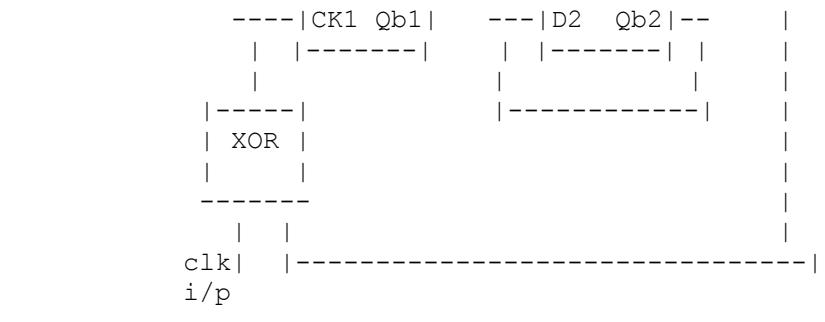


led should glow when q=0 and off when q=1
the choices are
a.ckt will funct as given
b.it wont funct as given
c.q cant drive ttl inverter
d.non of these
ANS: b

6)n nets are givenin how many ways can we model each
of the stuck at fault in n nets in single(one at a time) and w....
ANS. 2n and (3^n)-1

7) circiut with 2 d ffs was given





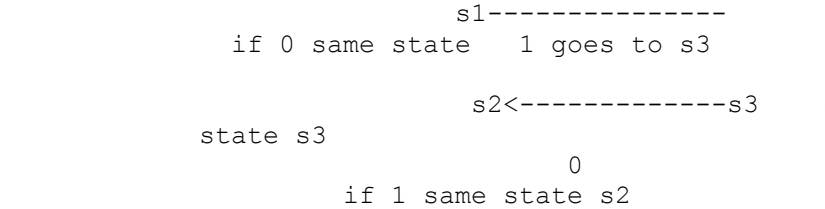
whaT is the relation between B and clk i/p?

Ans. b is 1/3 of clk i/p with 50% duty cycle.

8)3 dffs was given with common clk
setup time 3ns
hold time 1ns
clk to q delay 2ns find the maximum frequency of operation

ANS: 200MHz

9)fsm question there states given ques what is the
machne called



ans: MOORE M/C

10. consider a adder and multiplier
question ios some what like can both implemented in
same fsm or cant

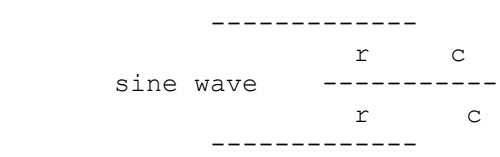
ANS: Yes they can implemented in same machine.

these are the 10 ques asked in digitl section and we are in a
situation to attend 2 section i attented analog part
and i am giving the ruogh idea of analog section

****
Analog part 10questions

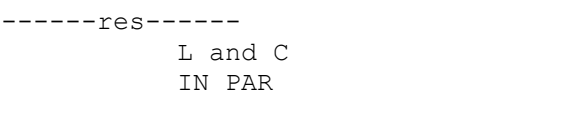
1. input is sine wave
what is the op in the middle of serially connected 2

rc parallel ckts



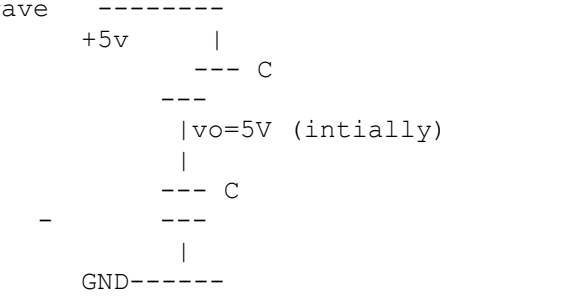
a. cos wave
b.sine wave with 0 phase shift
etc
ANS: b

2.square wave as input given to the ckt



ANS: SINE WAVE.

3.in 5v wave

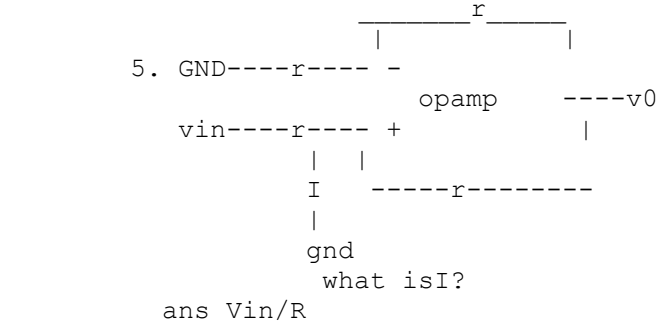


to the circuit in which two cap are connected in series
what is the op at the vo at time T
intially it was at 5v

ans: may be 5V. (CHECK IT MAY BE 2.5 V)

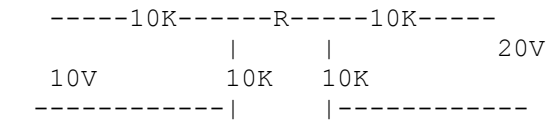
4.darlinton pair was given with beta as ct gain of each trans
what is the overall gain
ans: (beta+1)^2

5.



what isI?
ans Vin/R

6.

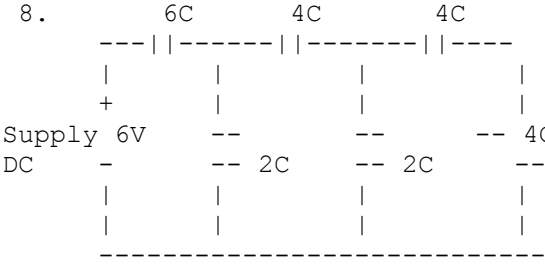


7. IF THE INPUT SGL IS 95khz and it is sampled at 120samples per sec the at what freq wil the fft opt fundemental freq will come

totally there were 10 questions in analog ssection also

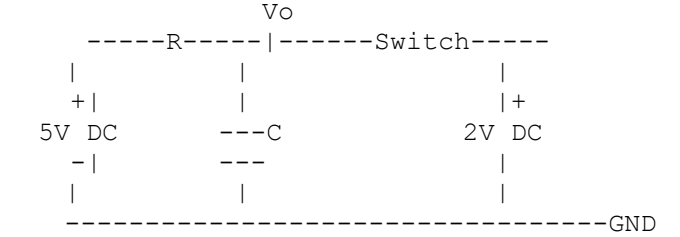
the numbering is not right

8.



find out Vo=? (caps vaule may be changed).
ANS was 0.75V but here i think cap. values changed, so calculate urself for currect answer.

9.



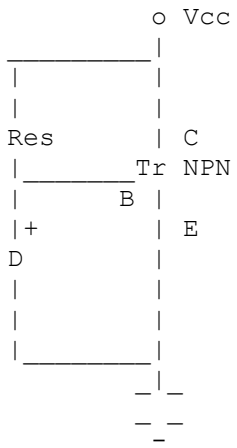
Switch is open at t=0, what is the value of Vo at t=infinity. Ans. 5V

10. one Qs on current mirror. there are n current mirrors are connected in series, you have to find out
the condition for whitch current mirrors will be in linear region.
(a)Vt
(b)Vt+ deta V
(c)n*Vt+ (n-1)* delta V
(d)n(Vt+delta V)

ANS: c

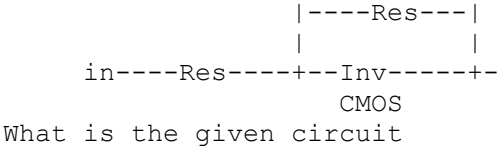
paper of TI 1999
Hard ware part only. There was one part of reasoning and there was separate paper for software persons.

1.



Find the current I delivered by the battery.

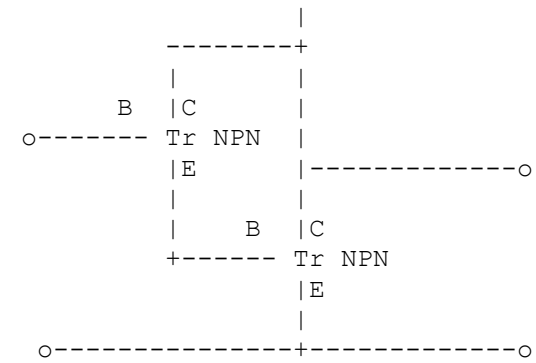
2.



What is the given circuit
a) Latch b)Amplifier c)Schmitt trigger. d)

3. The total no of flip flop required for N stage sequential circuit
N N-1 N
a)2 b)2 c) Log N d) 2 -1

4.

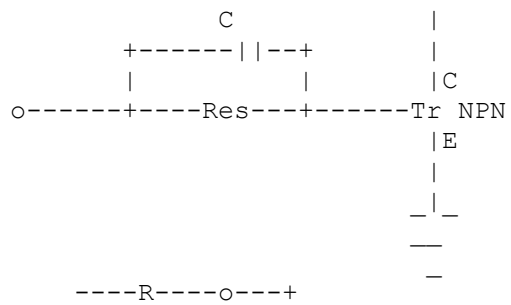


the gain of the circuit is
a) beta square b)beta + 1 c) (beta+1) ka square d)

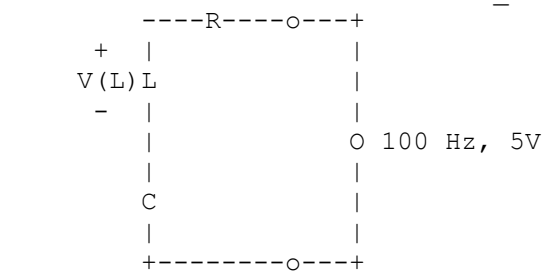
5. If the o/p and i/p are related by y=k(x square) and i/p is a sum of 2 waveforms
then the modulation scheme is
a) FM b)AM c)PM and d)None
Ans. B

6.Function of C in the circuit below is
a) Improve switching b)dc coupling c) ac coupling d) None

o



7.



if the ckt is at resonance and V(L)= (constant) V (given)
the value of V(R) and V(C) is
a)100V,5V b)-100V,5V c)5V,5V
   (Use V(L)=5 /_100 and V(C)=5/_-100, V(R)=5V

8. Minimize the K-map

	A'B'	A'B	AB	AB'
c'	1	X	0	1
c	1	X	0	1

a) A'B' b) A'+B' c)B' d)A'+B'+C'

9. IF the rate of removal of elements in a queue containing N elements is
proportional to the no of elements already existing in the queue at that
instant then the no. of elements----
a)decrease linearly b)Exponetialy decrease b) Logarithmcally

10. One question on CMOS ckt.
11. Two question on OP-AMP.

THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple
ckt.one ^M
>transistor is diode equivalent. & asked the o/p across the 2 nd
transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
> Emitter^M
>---R-------transistorbase| --^M
> | ---^M
> collector^M

> in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
> in fig.capacitor is used for in this ckt:^M
>^M
>^M
> ans:a.speedupb.active bypass c.decoupling^M
> 4.^M
>^M
> -----R------I----------o/p^M
> |___R____ |^M
> in above r is resistence.I is cmos inverter.^M
> then ckt is used for:^M
>^M
>^M
> a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
> 5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
> amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
> from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
> 6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M
> inductor is given.R I C values are given & asked for^M
> voltages across resistence & capacitor.^M
> 7.^M
> ___ R_____^M
> | |^M
> ---R------OPAMP ----------^M

> |---^M
> R1 R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
> ground^M
>^M
>^M
> 8.asked for Vo at the o/p.it is like simple cmos realization that is
n ^M
>block is above^M
> & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
> 9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay
^M
>is 1 nanosec.^M
> A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
> a.updown^M
> b.up c. updown glitching like that (take care abt glitching word)^M
>^M
> 10.^M
>^M
>^M

> ----------------| subtractor|---------o/p^M
> |___HPF____|^M

>^M
> the ckt is LPF ,HPF or APF ?^M
>^M
> 11.in a queue at the no of elements removed is proportional to no of
^M
>elements in^M
> the queue.then no of elements in the queue:^M
> a.increases decreases exp or linearly(so these are the 4 options
given ^M
>choose 1 option)^M
> 12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the
^M
>fastest in the^M
> following implementations.^M
> ans we think ((AB)(CD))((EF)(GH))^M
> 13.with howmany 2:1 MUX u can for 8:1 MUX.answer is 7.^M
> 14. there are n states then ffs used are log n.^M
> 15.cube each side has r units resistence then the resistence across ^M
>diagonal of cube.^M
> 16.op amp connections asked for o/p^M
> the answer is (1+1/n)(v2-v1).check it out.practise this type of
model.^M
> 17.^M
> _____________ supply^M
> ---|__ ___|^M
> Ii >________ |___ Tranistot^M
> > _______Vo^M
> > _______Vo^M
> |^M
> |^M
> R |^M
> | | Io^M
> ground.^M
>^M
>^M
>^M
>^M
> asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
> a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
> 18.y=kxsquare. this is transfer function of a block with i/p x & o/p
^M
>y.if i/p is^M
> sum of a & b then o/p is :--^M
>^M
> a. AM b.FM c. PM^M
> 19.^M
> ------MULTIPLIER--- |^M

> | |^M
> _____R__|__OPAMP_____________________Vo^M

> ---^M
> |^M
> ground.^M
> v in = -Ez then o/p Vo =?^M
> answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
> is a.b;^M

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in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are
diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
----
c 1 x 0 0
1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```

what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

```
7.#define f(a,b) a+b
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```

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bye..
p.sreenivasa rao

_____

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> ATTACHMENT part 5 application/msword name=texas.doc

> ATTACHMENT part 6 application/octet-stream name=texas

Date:
Wed, 30 Dec 1998 19:30:34 +0500
From:
PVSAK Viswanadham Add to Address Book
Subject:
TI
Organization:
Computer Science Dept., Indian Institute of Technology, Kharagpur
To:
bkup

for TI aptitude test consist of all pictorial questions. ie in each question he will give 8 diagrams and ask to find the 9'th diagram in that sequence.
You go through RS Agarwal. These aptitude questins are very easy. Just pratice them. In RS Agarwal gothrough SERIES chapter. It is suffient. There are 35 aptitude questions. First 25 are very easy. Do these questions in just 15 or 20 minutes. Because last questions are very touch.

TECHNICAL TEST:
---

1)3 flipflops are connected so that after 0 to 5 count occured next number is zero. So what is the counter?
Ans: mod 6 counter

2)simplication of some boolean _expression which is simple.
Boolean _Expression is A+A'B.
Ans:A+B

3)Given inorder sequence and preorder sequence and asked to find out postorder sequence.

4)Some question on value of a static variable.

5) Given an interger in binary form,find the number of ones in that number without counting each bit.(This questin is not multiple choice question. This question carries more marks. So please take care for this question.)

6) 1-way set associative memory is called-----
a)direct b)something c)1-way set associative 4)something
Ans: c

7)Fastest IPC mechanism is

a)shared memory b)pipes c)named pipes d)semaphores
Ans:c

8)Some page references are given. You are asked to implement it with Least Frequently Used algorithm.

9)Some diagram is given.
Iam describinmg the diagram. A 2*1 MUX is given. The inputs are A,B. Output is C. C and A are tied together. What is the diagram.?

Ans:Latch.

****

This paper is for Electrical & Electronics students. There is separate test for computer Science Students. There are 20 questions.

1)Some circuit is given. Iam describing the circuit.
A resistor R & a capacitor C are connected in parallel.
To this circuit another circuit which is having a capacitor of capacity 2C & an impedence Z, is connected in series.
You are asked to find out the value of Z? Note that 2C & Z are connected in series.
a)Z=2C
b)Z=2L
c)Z=L/2
d)Z=2R

2)Some circuit which consist of only resistors R is given.
This is a repetative circuit. U have to find the effctive resistance of the entire circuit.
A)Rin=R
B)Rin=(5+sqrt(3))/7
C)Rin=(19+sqrt(3))/8
D)None.

3)Two wave forms are given. You are asked to write the cirsuit to get B(second wave form) from A(first wave form).

4)#define SUM(a,b) a+b

```
main()
{
a=2;
b=3;
x=SUM(a,b)*2;
printf("x=%d\n",x);
}
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
```

```
}

main()
{
static int i=0;
number(i);
}
```
Ans: I don't know.

6)Some circuit is given. I can't describe the circuit. There are
3 resistors,3 capacitors & one inverter.. The question is
What is the value of the frequency such that the circuit oscillates.
A)f=RC
B)f=sqrt(3)/(Pi*R*C)
C)f=1/(Pi*R*C)
D)something
Ans:I don't know the answer.

7)Question on flipflop. So gothrough all flipflops.

8)There are 5 questions on Nmos & Pmos circuits.

****
This Paper is for Computer Science Students. THis paper is
very easy. You can definitely do it in one hour.
****

(1) The fastest memory is
(i) DRAM, (ii) ROM, (iii) SRAM, (iv) Main memory
Ans : SRAM

(2) Programing exceptions are
(i) Asynchronous, (ii) Synchronous, (iii) None
Ans : Asynchronous

(3) DSP which architecture is used
(i) MIMD, (ii) SIMD, (iii) Nueman, (iv) Harvard Architecture
Ans : Harvard Architecture

(4) C prog. for searching for an element in linked list

```
(5) main()
{
unsigned char i;
int sum;

for(i=0; i<300; i++)
sum+ = i;
printf("\nSum = %d\n", sum);
}
```

Ans : infinite loop

```
(6) void fn(int *p)
{
static int val = 100;
p = &val;
```

```
}

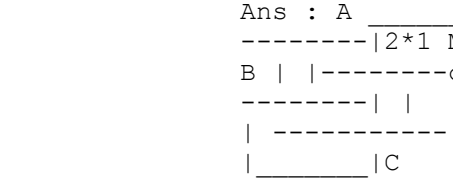
main()
{
int i=10;
printf("i=%d\n", i);
fn(&i);
printf("i=%d\n", i);
}
```

Ans : i=10 i=10

(7) int a[10[15];
char b[10[15];
(a) location g a[3][4], if base location g a[0][0] is ox1000
(b) location g b[3][4], if base location g b[0][0] is ox2000
int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans : A


B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

```
(i) x = x+y;
y = x-y;
x = x-y;

(ii) x = x^y;
y = x^y;
x = x^y;
```

(11) Count no of 1's in a word without using bit by bit.
(This question carries more marks. It is not a multiple choice question.)

```
(12) Code 1 :
for(i=0; i<1000; i++)
for(j=0; j<100; j++)
x = y;

Code 2 :
for(i=0; i<100; i++)
for(j=0; j<1000; j++)
x = y;
```

Which code will execute faster
(i) Code 1 and Code 2 are of same speed,
(ii) Code 1,
(iii) Code 2,
(iv) None.

Ans : Code 2

(13) main()
{
int a[10] = {1, 2, 3, ..., 10}, i, x=10, temp;

for(i=0; i temp = a[i];
a[i] = a[x-i-1];
a[x-i-1] = temp;
}
(i) All contents of array a are reversed
(ii) Only some portions are altered
(iii) Remains same
(iv) None

Ans : (iii)

(14) An array is stored in row major order. The memory capacity is 30 MB. And in unix system demand paging is used. Which one will give more page faults?

#define V_L_I 10000
int i, j, array[V_L_I][V_L_I];

Code 1 :
array[i][j] = 1;

Code 1 :
for(j=0; j for(i=0; i array[i][j] = 1;

Ans : Code 2

(15) In C which parameter passing technique is used?
(i) call by value,
(ii) call by reference,
(iii) both

Ans : call by value

(16) A circuit is given with 2 exclusive OR gates whose boolean _expression will be y = '(AB) + AB
(' indicates bar)

(17) main()
{
int i = 1;
fork();
fork();
printf("\ni = %d\n", i+1);
}

Ans : 4 printfs will occur and i = 2

(18) Compute the complexity of Binary search.
Ans : O(lg n) (Answer in detail. This is not a multiple choice question.
It carries more marks.)

(19) Write _expression for the tree graph :
Ans : ((a-b) + c*d)/x

(20) # define MAX(a, b) a>b ? a:b
```
main()
{
int m, n;
m = 3 + MAX(2, 3);
n = 2 * MAX(3, 2);
printf("m = %d, n = %d\n", m, n)
}
```

Ans : m=2, n=3
paper of texas instruments.
Technical + aptitude + interview.
* Questions on c
here i am not strictly following syntax it is just to show what was asked.
1 #define sum(a,b) a+b
value of sum(2,3)*2 Ans:8

2. a=5,b=6
swap(&a,b);
This function is written to swap a and b
find value of a and b . Ans 6,6

3.
```
function()
{
static int a=0;
a=a+1;
return a;
}
main()
{
function();
function();
function();
printf a;
}
```
final value of a ? Ans : a=3. static initializes once.
4.Write two prog. to swap a & b without using temp variable.
5.
```
unsigned char i;
int sum=0;
for(i=0;i<300;i++)
sum=sum+i;
printf(sum);
```

Ans:Program will held in infinite loop b/c i can not exceed 255.

6.
five questions on MOSFETS.
four were having single mosfets.
questions were simple.as i told u in Gwalior.
just on the funda that it will conduct if Vg-Vs > Vt .

In one question output at drain was to be calculated while o/p was initially
charged to 5v and to the gate 5v were applied.

In one question output at drain was to be calculated while o/p was initially
charged to 5v and the gate was shorted to drain.
7.
Clear the concept of settling time , hold time and other times. 3 ques on that.
like values of various delays were given and max frequency at which the circuit can work
hint : 1/sum of all delays . In our case ans was 200 Mhz.

8. An input and output waveform was given and circuit was to be designed with the use of one
delay.
Ans : exor gate in which second input is first input with a delay.

9. A question to determine sequence of counter. Don't get puzzled it was a
tough question.

10.The output and input of a inverter is connected by three RC stages in between of each stage two amplifiers with poles at imaginary axis were connected.
Hint : The poles at imaginary axis will create extra 180 phase shift thus the
circuit will oscillate and calculate the frequency of operation.

11. A series of infinite connected rc circuit and overall input resistance is
calculated.(question of 12 class)

1: given an expression tree and asked us to write the in fix of that expression

four choices

2:
global variables in different files are

a:at compiletime
b) loading time
c) linking time
d)execution time

3)size of(int)

a) always 2 bytes
b) depends on compiler that is being used
c) always 32 bits
d) can't tell

4)which one will over flow given two programs
2

prog 1:

```
main()
{
int fact;
long int x;
fact=factoral(x);

}
```

prog2:

```
main()
{
        int fact=0
for(i=1;i<=n;i++)
        fact=fact*i;

}
```

```
int factorial(long int x)
{

if(x>1) return(x*factorial(x-1);
}
```

a) program 1;
b) program 2;
c) both 1 &2
d) none

}

5) variables of fuction call are allocated in
a) registers and stack
b) registers and heap
c) stack and heap
d)

6)

avg and worst case time of sorted binary tree

7) data structure used for proority queue
a) linked list b) double linkedd list c)array d) tree

8)

```
 main(){
char str[5]="hello";
if(str==NULL) printf("string null");
else printf("string not null");
}
```

what is out put of the program?
a) string is null b) string is not null c) error in program d) it executes but print nothing

9)there are 0ne 5 pipe line and another 12 pipe line sates are there and flushed time taken to execute five instructions

a) 10,17
b) 9,16
c)25,144
d)

10)

for hashing which is best on terms of buckets
a)100 b)50 c)21 d)32 ans 32

11)

```
void f(int value){
for (i=0;i<16;i++){
if(value &0x8000>>1) printf("1")
else printf("0");
}
}
```

what is printed?
a) bineray value of argument b)bcd value c) hex value d) octal value

12)

```
void f(int *p){
static val=100;
val=&p;
}
main(){
int a=10;
printf("%d ",a);
f(&a);
printf("%d ",a);
}
```

what will be out put?
a)10,10

13)

```
struck a{
int x;
float y;
char c[10];
}
union b{
int x;
float y;
char c[10];
}
```

which is true?

a) size of(a)!=sizeof(b);
b)
c)
d)

14)

```
# define f(a,b) a+b
#defiune g(c,d) c*d
```

find valueof f(4,g(5,6))
a)26 b)51 c) d)

15)

find avg access time of cache
a)tc*h+(1-h)*tm b)tcH+tmH

c) d) tc is time to access cache tm is time to access when miss occure

16)

```
main()
{
char a[10]="hello";
strcpy(a,'\0');
printf("%s",a);
}
```

out put of the program?
a) string is null b) string is not null c) program error d)

17)

simplyfy k map

1 x x 0
1 x 0 1

18)

```
int f(int a)
{
a=+b;

//some stuff


}

main()
{
x=fn(a);
y=&fn;
```

what are x & y types
a) x is int y is pointer to afunction which takes integer value

19) char a[5][15];
int b[5][15];
address of a 0x1000 and b is 0x2000 find address of a[3][4] and b[3][4]
assume char is 8 bits and int is 32 bits

a) b) c) d)

there are 20 questions all in techinical paper and 36 questions in appititude test

in appititude thay have given all diagrams and asked to find what comes next

thay are quite easy and i hope if u practice r.s aggraval u can do it easily

for tecnical thay have given 1 hr for 20 questions and for not technical thay have given only 40 min and 36 questions

this is the paper i have right now

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c 1 x 0 0
   1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

```
4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```

what's the purpose of the program

a. its output is hex representation of i
b. bcd

c. binary
d. decimal

7.
```
#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.
```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance

is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input
because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
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which is correct .
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fork();
fork();
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}

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a.3
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d. 8

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```
for (j=0;j<16;j++)
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```

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```
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```

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